

CS355: Web Technologies - Lab

Lab 5: PHP

08 , 09 April 2025

1. PHP (Hypertext Preprocessor)

PHP is a server-side scripting language primarily used for web development. It's known for its simplicity, flexibility, and compatibility with various databases, servers, and operating systems.

Key features:

- It runs on the server and generates HTML that is sent to the client's browser.
- Could be embedded in HTML, making it easy to create dynamic web pages.
- PHP is often used with a database to build dynamic, data-driven websites.

2. PHP Basic Syntax

1. PHP code is written **between special tags**

```
<? php  
// Your PHP code goes here  
?>
```

2. Each PHP statement ends with a **semicolon (;)**

3. Variables in PHP start with a **dollar sign (\$)**

```
$name = "Ahmad" ; //This is a String  
$age = 25; //This is an integer  
$gpa = 98.5; //This is a float  
$available = true; //This is a Boolean  
$color = [ "red" , "blue" , "green" ]; //This is an array
```

4. Echo/Print used to display output:

```
echo "Hello!" ;  
print "World!" ;
```

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5. The **Control Structures: If/Else**

```
if ( $age >= 18 ) {  
    echo "Adult!";  
} else {  
    echo "Minor!";  
}
```

6. **Loops**

```
// For loop  
for ( $i = 0; $i < 5; $i++ ) {  
    echo $i;  
}  
  
// While loop  
$i = 0;  
while ( $i < 5 ) {  
    echo $i;  
    $i++;  
}
```

7. **Functions**

```
function greet ( $name ) {  
    return "Hello, " . $name;  
}  
echo greet ( "Alice" );
```

8. **String Functions**

```
$name = "Ahmad" ;  
echo strlen($name); // Get string length, Output: 5  
echo strtoupper($name); // Change case, Output: AHMAD  
echo strtolower($name); // Change case, Output: ahmad  
echo substr($name , 2, 3); // Returns a portion of a string, Output: mad  
echo strpos($name , "hm"); // Finds the position of the first occurrence of a substring, Output: 1  
echo str_replace ( "Java" , "PHP" , "I love Java." ); // Replaces all occurrences of a substring with another, Output: I love PHP.  
echo trim($name); // Removes whitespace (or other characters) from the beginning and end of a string
```

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3. HTML Forms with PHP

HTML creates the form (what users fill in). while the **PHP processes the form data** (like saving, validating, displaying, etc.).

The **HTML form** must declare 2 attributes to use PHP:

1. The PHP file that will **handle the form**, using the “action” attribute.
2. The method that will send the data from the HTML to PHP:
 - **GET** method appends form data to the URL as query parameters. Used for search, filters, non-sensitive data.
 - **POST** method sends form data in the body of the HTTP request, hidden from URL. Used for login, forms, sensitive info.

Example

1. HTML Form:

```
<!-- form.html -->
<form action= "process.php" method= "post">

    <label>Name:</label>
    <input type="text" name= "username" > <br>
    <label>Email:</label>
    <input type= "email" name= "email" > <br>
    <input type= "submit" value= "Submit" >
</form>
```

2. PHP Script to Process the Form:

```
<!-- process.php-->
<? php
if ( $_SERVER [ "REQUEST_METHOD" ] == "POST" ) {
    $name = $_POST[ 'username' ];
    $email = $_POST[ 'email' ];

    echo "Name: " . htmlspecialchars( $name ) . "<br>";
    echo "Email: " . htmlspecialchars( $email );
}
?>
```

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4. Running a PHP form in NetBeans

A. Install a local web server: XAMPP or WAMP

Download XAMPP: <https://www.apachefriends.org/index.html>

What is XAMPP / WAMP?

PHP is a **server-side language**, which means it needs a **web server** and a **PHP interpreter** to run.

Instead of buying a real web server, **XAMPP** and **WAMP** tools simulate a server **on your own PC**.

They are **software packages** used to run a **local web server** on your computer, they include:

1. PHP: A Language interpreter for PHP code
2. Apache: A Web server that runs your PHP scripts
3. MySQL: A Database engine for storing data
4. phpMyAdmin: A Web tool to manage MySQL (with GUI)

B. Start Apache in XAMPP

1. Open XAMPP Control Panel
2. Click **Start** next to **Apache**
3. Make sure it's running (green highlight)
4. This sets up your **local server** (usually at <http://localhost/>)

C. Set Up a PHP Project in NetBeans

1. Open NetBeans
2. Go to **File > New Project**
3. Choose: **Category:** PHP , **Project:** PHP Application
4. Click **Next**
5. Browse and select your **project folder** (e.g., inside C:\xampp\htdocs\myform)
6. Set the **Run Configuration:** *Local Web Site*, Project URL: <http://localhost/myform>

D. Add Your HTML + PHP Files

E. Run the Project

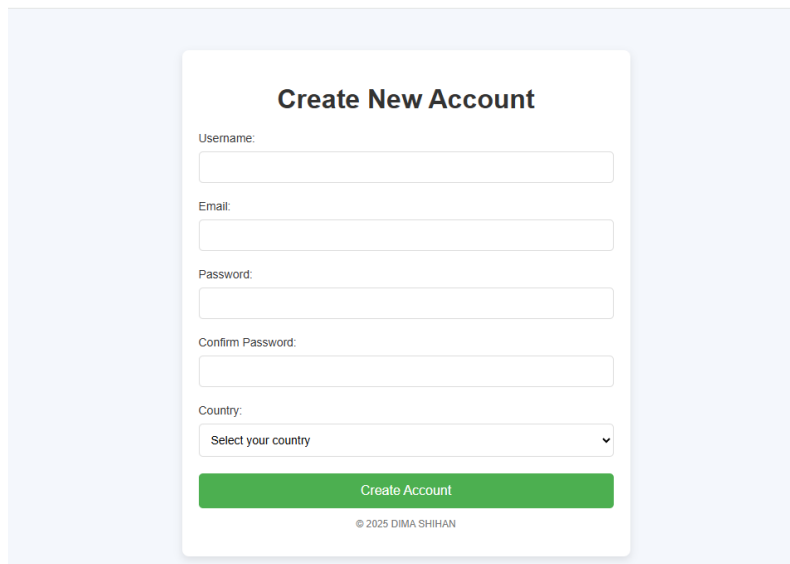
1. Right-click the project name in NetBeans
2. Click **Run**
3. Your browser should open: <http://localhost/myform/form.html>

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5. Task

Create a simple PHP project , as follows:

1. An HTML form that asks the user to enter their information: Username [type text], Email [type email], Password [type password] and Country [select list].



2. Add proper styling to the form:
 - Place the form inside an HTML class

Note: To get the box effect use the following,

 - margin: 50px auto
 - background-color: #ffffff;
 - border-radius: 8px;
3. When submit button is clicked, the HTML form sends the information to the PHP file [index.php]
4. PHP should validate the data (no empty fields, year is a positive number) show a welcome sentence contains as in the screenshot.

